SOLAR PV AS A NON-WIRES ALTERNATIVE IN ONTARIO'S ELECTRICITY PLANNING FRAMEWORK

Policy & Valuation (PV) Research Report

Abstract

This report evaluates the economic, regulatory, and system-planning conditions shaping the role of solar PV as a non-wires alternative within Ontario's evolving electricity grid. It synthesizes leading technical and policy evidence to assess where PV-enabled DER portfolios can deliver dependable capacity, defer infrastructure, and support long-term grid modernization.

Andrew Facciolo

Andrew.fach1@gmail.com

Executive Summary

Ontario's electricity system is entering a period of unprecedented structural change driven by electrification, rapid load growth, aging infrastructure, and accelerating commitments to clean energy. Distributed energy resources (DERs), and solar photovoltaic (PV) in particular, are now positioned to play a materially greater role in how the province manages capacity needs, defers infrastructure investment, and strengthens long-term system reliability.

This report synthesizes evidence from a selection of trusted technical, regulatory, and economic sources, including the NREL Annual Technology Baseline (2024), the CanREA Market Outlook (2025), the Smart Prosperity Institute (2021), the IESO–ICF York Region NWA Demonstration (2024), the OEB's Framework for Energy Innovation (2023), and the 2025 GTA North IRRP. Across these sources, several consistent conclusions emerge.

First, PV is becoming one of Ontario's most economically competitive resources. Costs continue to decline across utility-scale, commercial, and residential systems, with utility-scale LCOE approaching $20–26/MWh by mid-century. Canadian market projections show similar downward trajectories, placing solar among the lowest-cost sources of new electricity in the province. These cost declines align with significant national growth expectations, with PV capacity projected to expand from roughly 3 GW today to 8 GW by 2030 and 20 GW by 2050.

Second, DERs—including PV-enabled portfolios—have proven operational value in Ontario. The York Region NWA Demonstration demonstrated that DER portfolios can be reliably dispatched for peak relief, achieving 83% availability, 85% performance accuracy, and oversubscription in successive Local Capacity Auctions. These outcomes confirm that properly coordinated DERs can provide dependable, location-specific capacity under real operating conditions.

Third, Ontario's regulatory framework is now aligned with large-scale DER deployment. The Ontario Energy Board's Framework for Energy Innovation (FEI) introduces the province's first standardized Benefit-Cost Analysis (BCA) Framework, clarifies cost recovery for DER-related expenditures, and enables pilot incentive mechanisms to counteract historical capital-bias in utility remuneration. In parallel, the IESO's NWA Screening Tool and updated IRRP methodology formally integrate DERs into long-term regional planning.

Fourth, regional electrification pressures create clear opportunities for PV-based NWAs. The 2025 GTA North IRRP forecasts extreme-weather summer peaks rising from 2,600 MW to 4,200 MW by 2043, and winter peaks rising from 1,900 MW to 5,700 MW. While legacy FIT and microFIT solar currently reduce peak demand by only 14–33 MW, DER portfolios are expected to provide 60–80 MW of deferrable capacity across screened-in subregions such as Northern York, Markham, and Parkway–Buttonville. These constraints reveal precisely where incremental PV and DER resources can produce measurable planning value.

Finally, PV contributes a quantifiable share of dependable capacity. The DER Potential Study assigns PV a 28% summer peak-coincidence factor, and hybridization analysis identifies a 15–30% dependable-capacity contribution for standalone PV depending on tilt and siting. These

values, while modest compared to storage or demand response, remain significant when aggregated across thousands of rooftops and integrated into local flexibility portfolios.

Across all sources, the evidence points to a clear inflection point: Ontario now has the economic conditions, operational proof, regulatory tools, and regional capacity needs required to integrate PV as a meaningful component of its non-wires alternative (NWA) strategy. PV is not a replacement for major transmission reinforcements, particularly under emerging winter peaks, but it is a scalable, increasingly low-cost, and reliably quantifiable contributor to summer-peaking needs and a foundational resource in broader DER portfolios.

For Deloitte and executive decision-makers, the implication is direct: **PV represents a credible, evidence-backed, and near-term NWA option that can support cost-deferral, emissions-reduction, and long-term system modernization objectives.** When paired with storage and demand response, PV becomes an important building block in a hybrid wires + non-wires planning strategy capable of enhancing reliability while reducing long-term capital outlay.

Ontario is no longer in the exploratory phase. The province now possesses a maturing policy framework, validated operational models, and clear regional needs that justify systematic integration of PV into future capacity planning and DER procurement strategies.

While emissions and greenhouse gas (GHG) impacts are not explored in depth in this report, they remain a significant global driver of PV deployment and a key motivation for assessing PV's long-term system value (Smart Prosperity Institute, 2021, L1_03)

---

1. Introduction

Canada's electricity system is undergoing a fundamental transformation driven by decarbonization policy, electrification of end uses, and rapid technological change in renewable generation. Solar photovoltaic (PV) resources, once marginal contributors to the national electricity mix, have now emerged as a key pillar of long-term supply planning. PV currently makes up only a small portion of Canada's electricity—about 3.3 GW of installed capacity in 2019, or roughly 0.6% of national generation. Yet its growth trajectory is one of the most rapid in the non-emitting sector, positioning solar to become a materially more influential resource over the coming decades (Smart Prosperity Institute, 2021, L1_03). According to the Canada Energy Regulator's (CER) Evolving Scenario, national PV capacity is expected to increase to 8 GW by 2030 and 20 GW by 2050, reflecting more than a sixfold increase from the 2020 baseline (Smart Prosperity Institute, 2021, L1_03). These projections align with Canada's broader net-zero ambition and rising provincial commitments to clean electricity expansion. Across all DER classes identified in the Dunsky / IESO DER Potential Study (2022), PV accounts for approximately 80% of Ontario's behind-the-meter technical potential**,** making it the foundational resource driving the province's distributed resource landscape.

Ontario remains the dominant jurisdiction for PV deployment in Canada, historically accounting for 94% of installed PV capacity in 2019 due to earlier feed-in tariff (FIT) programs that stimulated market development (Smart Prosperity Institute, 2021, L1_03). While these legacy contracts are now expiring, Ontario's long-term decarbonization and electrification trends, especially in the Greater Toronto Area (GTA) are placing new emphasis on PV, distributed energy resources (DERs), and local grid flexibility.

Recent regional planning documents, including the 2025 Integrated Regional Resource Plan (IRRP) for GTA North, forecast significant load growth, with extreme-weather coincident summer peaks rising from 2,600 MW in 2024 to 4,200 MW by 2043, and winter peaks expanding even further to 5,700 MW by 2043 due to heating electrification (Independent Electricity System Operator [IESO], 2025, L2_03). These rapidly increasing peaks highlight the growing strategic value of PV generation and distributed flexibility resources within Ontario's system planning framework. Although PV's present-day contribution to regional peak reduction is modest—declining from roughly 33 MW today to 14–15 MW as legacy FIT and microFIT contracts expire—it remains a structurally relevant component of Ontario's load-modifying resources and a foundation for future DER-based flexibility.

Technological cost declines reinforce this trajectory. The U.S-based NREL Annual Technology Baseline (ATB) 2024, which provides detailed projections for utility-scale, commercial, and residential PV systems under the "Class 5 Moderate" case, shows continuous reductions in capital expenditures (CapEx) and levelized cost of energy (LCOE). For example, utility-scale PV CapEx decreases from $1,611/kW in 2023 to $683/kW by 2050, with LCOE falling from $49/MWh to $20/MWh over the same period (NREL, 2024, L1_01). Residential and commercial systems follow similar downward trends, contributing to PV's growing competitiveness relative to fossil-based generation. These cost trajectories align with Canadian national expectations, that by the mid-2020s, the LCOE of new solar and onshore wind will fall below that of natural-gas generation (Smart Prosperity Institute, 2021, L1_03), further cementing PV's role in future electricity supply portfolios.

Yet PV's value extends beyond bulk electricity supply. In high-growth constrained regions such as York Region, the integration of PV and DERs intersects directly with system planning and NWAs. The York Region NWA Demonstration, conducted jointly by the Independent Electricity System Operator (IESO) and Alectra Utilities, empirically tested whether DERs—including PV, demand response, and energy storage—could provide reliable distribution-level capacity (IESO & ICF, 2024, L2_01). Across 2021–2022, the local capacity auctions procured 10 MW and 15 MW, respectively, with the auctions oversubscribed by 1.9× and 1.5×, and DERs delivering 366 MWh of reliable peak-support energy with 83% availability and 85% performance accuracy (IESO & ICF, 2024, L2_01). These results confirm that locally sited DERs, including PV-enabled resources, can function as dependable components of grid planning when properly coordinated through a distribution system operator (DSO) model.

Regulatory modernization is evolving in parallel. The Ontario Energy Board's FEI introduces the province's first standardized Benefit-Cost Analysis (BCA) Framework for evaluating DERs as NWAs in utility rate applications (Ontario Energy Board [OEB], 2023, L2_02). It emphasizes that "costs should follow benefits" and that DER valuation must account for distribution-service impacts, DER-host impacts, transmission-service impacts, and broader system effects. The framework also outlines new pathways for utility incentives to adopt third-party DER solutions, addressing misaligned incentives that have historically favoured traditional capital investments over non-wires alternatives (OEB, 2023, L2_02).

Taken together, these developments—dramatic PV cost reductions, accelerating electrification-driven load growth, empirical NWA performance data, and modernizing regulatory structures—underscore the growing need to understand the comparative value of PV and DERs within the future Ontario electricity system. This report synthesizes technological, economic, regulatory, and system-planning evidence to evaluate PV's evolving role in electricity grids, with particular attention to how PV and DERs can contribute to capacity deferral, reliability, economic efficiency, and long-term electrification objectives.

---

2. Literature Review

2.1 Declining Costs and Technological Maturation of Solar PV

Recent literature consistently highlights the rapid decline in solar PV costs and the maturation of PV technologies, both globally and within the Canadian context. According to NREL's Annual Technology Baseline (ATB) 2024, capital costs for PV systems continue to fall across utility-scale, commercial, and residential segments. Under the Class 5 Moderate scenario—representative of North American mid-cost conditions—utility-scale PV capital expenditures drop from $1,611/kW in 2023 to $1,492/kW in 2025, and further to $895/kW in 2035, reaching $683/kW by 2050 (NREL, 2024, L1_01). Correspondingly, utility-scale LCOE declines from $49/MWh in 2023 to $20/MWh by 2050, reflecting both cost reductions and modest improvements in capacity factor from 26% to 30%. Commercial and residential systems follow similar trajectories, with residential LCOE declining from $140/MWh (2023) to $62/MWh (2050) (NREL, 2024, L1_01). To illustrate these cost trends, Figures 1 and 2 present the U.S. NREL ATB projections for utility-scale, residential, and commercial PV capital expenditures and levelized cost of energy (LCOE) under the Moderate Class 5 scenario. These visuals provide a clear view of the long-run downward trajectory that underpins the literature's consensus on continued PV competitiveness.

Figure 1

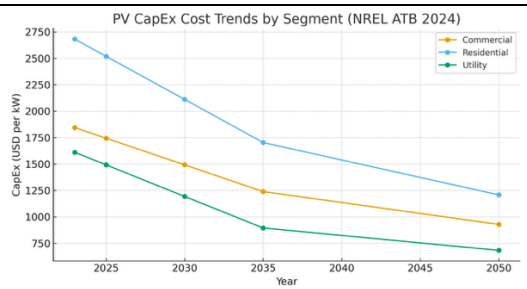
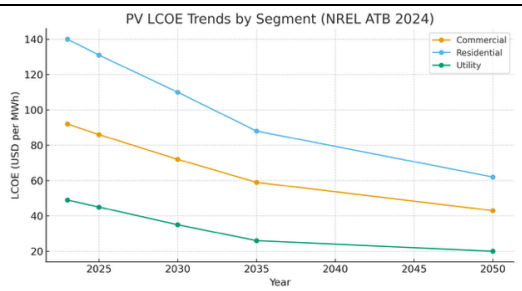


Figure 2



These cost reductions are aligned with broader Canadian projections. Smart Prosperity Institute notes that by 2026, the levelized cost of new wind and solar will fall below the cost of new gas generation, cementing solar as a competitive contributor to future electricity portfolios (Smart Prosperity Institute, 2021, L1_03). Canadian market dynamics are reinforcing this shift. Although installed PV capacity was a modest 3.3 GW in 2019, it's projected to grow rapidly—according to the CER's evolving scenario (Smart Prosperity Institute, 2021, L1_03). This transformation aligns with global PV cost declines and technology scaling. Figures 3 and 4 below provide a solid comparative view of Canadian CapEx and LCOE compared to U.S NREL ATB 2024 trends and is further expanded in section 3.1 of this report.

Figure 3

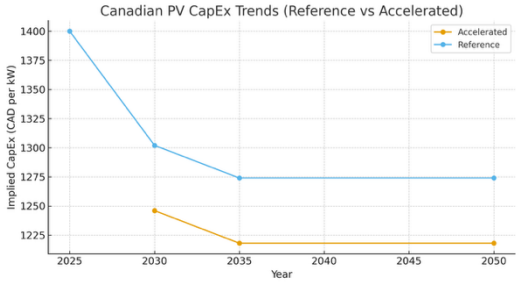
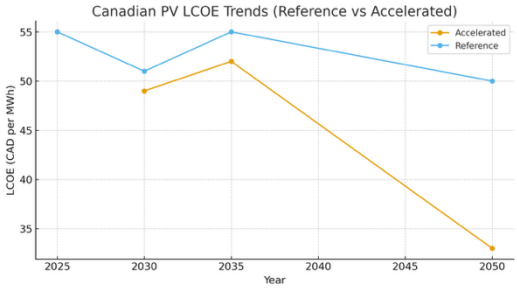


Figure 4



2.2 Geographic Distribution and Policy Legacy in Canada

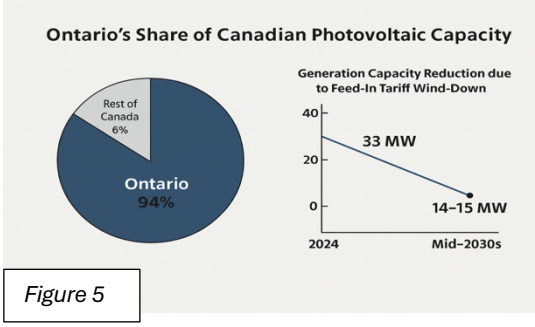


Figure 5

Ontario has historically dominated PV installation in Canada, accounting for 94% of national solar capacity in 2019 (Smart Prosperity Institute, 2021, L1_03). This concentration reflects legacy procurement programs such as the Green Energy and Economy Act's feed-in tariff (FIT) regime, which catalyzed early deployment but now limits incremental growth as contracts expire. The 2025 GTA North IRRP confirms this trend: distributed generation (mostly FIT/microFIT PV) reduces regional peak demand by 33 MW in 2024, but this value declines to 14–15 MW by the mid-2030s as contracts reach end-of-life (IESO, 2025, L2_03). Despite this modest contribution—less than 1% of extreme-weather coincident peak—the IRRP includes DG explicitly in its load-modifying resource forecasts, integrating PV's limited but measurable effect on regional planning (IESO, 2025, L2_03). As these legacy contracts wind down, their diminishing contribution to peak reduction underscores the importance of new solar and DER growth to maintain or enhance PV's system-level impact. The historical concentration of PV in Ontario and the diminishing peak-demand contribution of legacy FIT and microFIT systems are shown in Figure 5. Ontario's dominance—94% of national solar capacity in 2019—reflects the province's early procurement programs, while the declining peak reduction from 33 MW in 2024 to roughly 14–15 MW by the mid-2030s highlights the system-level implications of expiring contracts and the need for new DER growth.

2.3 Future Market Outlook: National and Provincial Perspectives

At the national level, PV growth is expected to be driven by decarbonization commitments and electrification across heating and transport sectors. The CER Evolving Scenario projects that solar will become one of Canada's fastest-growing resources, expanding more than sixfold between 2020 and 2050 (Smart Prosperity Institute, 2021, L1_03). Supporting evidence from Smart Prosperity Institute highlights the anticipated diversification of PV technologies: crystalline silicon, currently dominant, is expected to drop from 92% global market share (2014) to 45% by 2030 as thin-film and third-generation technologies expand (Smart Prosperity Institute, 2021, L1_03).

In Ontario, provincial long-term energy planning is increasingly integrating DERs and PV into system-wide considerations. For example, the Ontario Energy Board's Framework for Energy Innovation introduces a comprehensive Benefit-Cost Analysis Framework to evaluate DERs (including PV) as potential NWAs. The framework emphasizes a multi-layer valuation structure spanning distribution service impacts, transmission service impacts, DER-host implications, and general energy system benefits, addressing long-standing inconsistencies in DER cost-recovery treatment (OEB, 2023, L2_02).

Figure 6 summarizes these national and provincial dynamics. Nationally, PV deployment is expected to accelerate through mid-century alongside diversification from crystalline silicon toward thin-film and third-generation technologies. Provincially, Ontario's DER valuation framework introduces a structured, multi-layer approach for assessing PV across distribution, transmission, and broader system-benefit dimensions, reflecting its increasing integration into long-term planning.

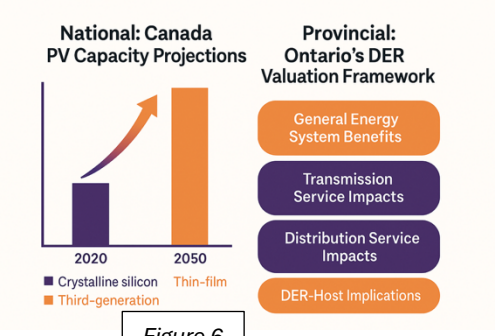
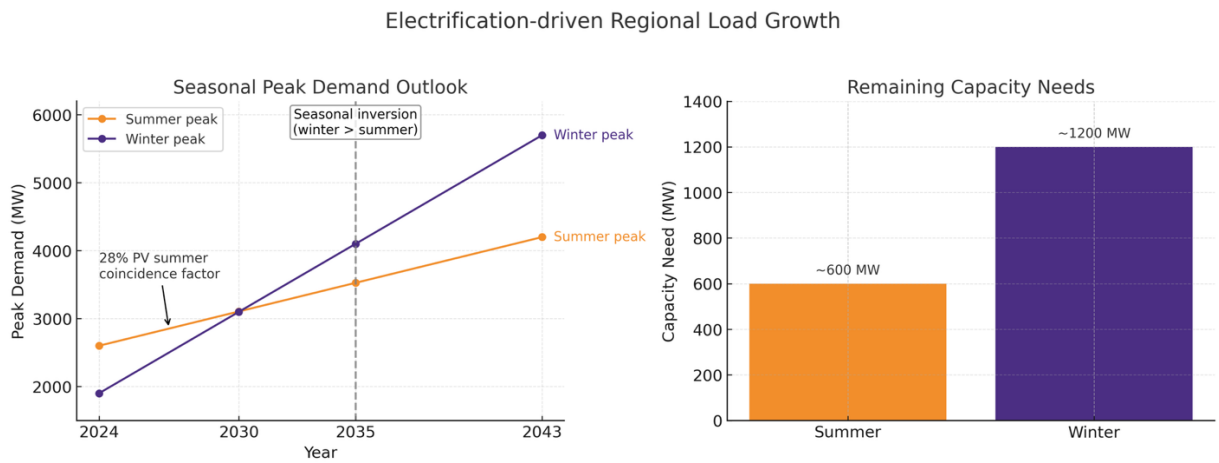


Figure 6

2.4 PV, Electrification, and Regional Load Growth

The most compelling literature on PV integration relates to the accelerating pace of electrification-driven load growth. The GTA North IRRP provides detailed 20-year forecasts showing that summer peak demand will grow from 2,600 MW (2024) to 4,200 MW (2043), while winter peaks—driven by heating electrification—grow even more substantially from 1,900 MW to 5,700 MW (IESO, 2025, L2_03). This seasonal inversion, expected to occur by the mid-2030s, positions winter reliability as a central concern for regional planners. This dynamic is consistent with the IESO/Dunsky DER Potential Study, which assigns rooftop PV a 28% summer peak-coincidence factor, indicating meaningful alignment between PV output and Ontario's highest summer demand hours. Against this backdrop, the IRRP identifies remaining unmet capacity needs of 600 MW (summer) and 1,200 MW (winter) after accounting for planned transmission and station upgrades. These residual needs represent the space where incremental DERs, including PV and demand response, could meaningfully defer or downsize infrastructure (IESO, 2025, L2_03). Figure 7 below visualizes Electrification-driven peak demand growth and remaining system needs.

Figure 7



2.5 DERs and Non-Wires Alternatives in the Literature

The strongest empirical foundation for DER-based NWAs in Ontario arises from the York Region NWA Demonstration, evaluated in 2024 by ICF (IESO & ICF, 2024, L2_01). Over two years (2021–2022), the demonstration procured 10 MW and 15 MW of local DER capacity through competitive auctions that were oversubscribed by 1.9× and 1.5×, respectively. DERs delivered 366 MWh across 15 activation events with 83% availability and 85% performance accuracy, demonstrating that aggregated DER portfolios can reliably provide peak capacity under a utility-administered market model (IESO & ICF, 2024, L2_01).

Critically, the evaluation quantified projected 2027 DER deferral value in $/MW-year terms (See Table 1).

Table 1

DER Deferral Value by Scenario ($/MW-Year)

Scenario	Year	DER Deferral Value ($/MW-yer)
2027 Base Case	2027	+391.000
2027 High Growth	2027	+382.000
2032 High Growth	2032	+515.000

(Source: IESO & ICF, 2024, L_2 01)

These values reflect avoided generation, transmission, and distribution capacity costs—components that align directly with the BCA categories defined by the OEB FEI (OEB, 2023, L2_02). The strong economic performance of DR-heavy portfolios in the York Region pilot provides empirical grounding for Ontario's pivot toward integrating DERs into Distribution System Plans (DSPs) and IRRPs (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02).

2.6 Regulatory Frameworks and Institutional Readiness

Ontario's emerging regulatory environment demonstrates increasing alignment with DER-integrated planning. The OEB FEI not only introduces standardized BCA methodology but also acknowledges structural disincentives that have historically discouraged utilities from adopting third-party DERs. To address this misalignment, the FEI outlines a suite of potential incentives—including shared savings mechanisms, scorecard-based incentives, and margin-on-payments options—that utilities may test in upcoming rate applications (OEB, 2023, L2_02). These mechanisms represent transitional steps toward a more comprehensive remuneration reform aligned with the province's electrification and energy transition objectives.

The IRRP process itself has become an institutional venue for DER integration. The GTA North IRRP repeatedly references the need for incremental eDSM, local DER portfolios, and targeted NWAs to manage 60–100 MW of deferrable capacity across Northern York, Markham, and Richmond Hill subregions (IESO, 2025, L2_03). While the IRRP does not reference the York Region NWA Demonstration by name, its methodology and screening outcomes (Section 7.2–7.5) reflect direct application of the IESO's new NWA Screening Tool and OEB's Non-Wires Solutions Guidelines and evidence that DER and PV integration are now embedded components of Ontario's planning philosophy (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02; IESO, 2025, L2_03).

---

3. PV Economics in Ontario's Electricity System

3.1 Capital Costs, LCOE, and Comparative Value

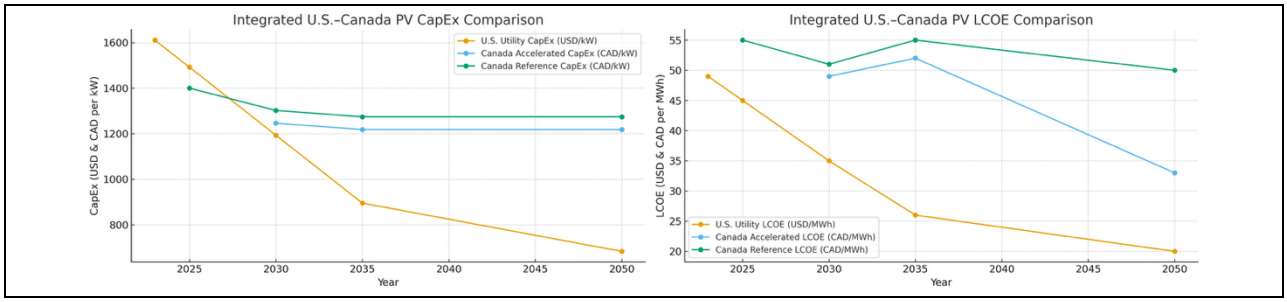
Solar PV economics in Ontario are increasingly shaped by falling capital costs, improved performance, and integration within broader electrification-driven load growth. As documented in the U.S NREL ATB 2024, utility-scale PV demonstrates substantial cost declines under the Class 5 Moderate trajectory. Capital costs decrease from $1,611/kW (2023) to $1,492/kW (2025), then reach $895/kW by 2035, and ultimately $683/kW by 2050, with LCOE falling from $49/MWh (2023) to $20/MWh (2050) (NREL, 2024, L1_01). These cost reductions are complemented by modest capacity factor improvements from 26–30%, reflecting both module efficiency gains and system-level optimizations. Commercial and residential PV experience similar but less pronounced declines, with residential LCOE dropping from $140/MWh to $62/MWh across the same time horizon (NREL, 2024, L1_01).

Canadian-specific projections reinforce these trends. The CanREA 2025 Outlook estimates provincial solar capital costs of 1,400 CAD/kW in 2025, with reductions of 7–11% by 2030 (Reference vs. Accelerated scenarios) and 9–13% by 2035; by 2050, capital expenditures stabilize around 1,218–1,274 CAD/kW, depending on policy ambition (Canadian Renewable Energy Association [CanREA] & Dunsky Energy + Climate, 2025, L1_02). Translating these capital trajectories into LCOE, CanREA projects Ontario solar costs around 55 CAD/MWh in 2025, 51 CAD/MWh by 2030, with a temporary rebound in 2035 due to federal ITC expiry, and declines accelerating thereafter to 33 CAD/MWh by 2050 under the Accelerated scenario (CanREA & Dunsky Energy + Climate, 2025, L1_02). These values are consistent with NREL ATB mid-case trajectories when exchange rate and market conditions are normalized.

Figures 8 and 9 below compare long-term U.S. and Canadian solar PV cost trajectories, illustrating consistent declines in both capital expenditures (CapEx) and levelized costs of energy (LCOE). U.S. ATB projections show steep utility-scale cost reductions through 2050, while Canadian Reference and

Accelerated scenarios demonstrate parallel but more moderated declines shaped by domestic policy and market conditions.

Figure 8 *Figure 9*



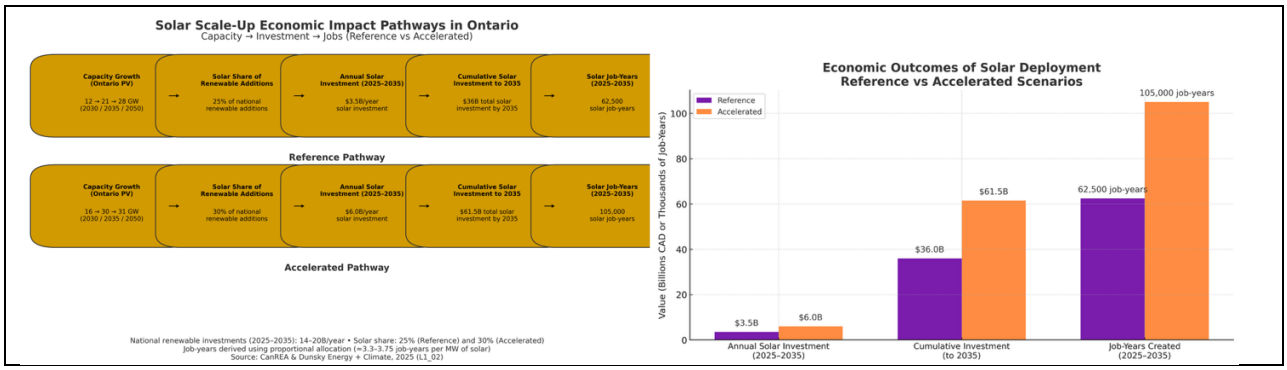
The industry's cost narrative is further bolstered by empirical evidence from Canada's earlier market development. Smart Prosperity Institute notes that PV's lifetime cost in 2016 was roughly 23 ¢/kWh, significantly higher than today's benchmarks, and projects that solar PV (paired with onshore wind) will undercut new gas-fired generation by 2026 (Smart Prosperity Institute, 2021, L1_03). Combined, these sources illustrate rapid cost convergence toward grid parity and strengthen the economic rationale for scaling PV in Ontario's decarbonized grid.

3.2 Investment Trajectories and Employment Impacts

The economic implications of scaling solar in Ontario extend beyond LCOE to include provincial and national investment flows and job creation. The CanREA 2025 Outlook estimates that solar installations will total 12 GW by 2030 (Reference) or 16 GW under the Accelerated pathway, rising to 21–30 GW by 2035 and 28–31 GW by 2050 (CanREA & Dunsky Energy + Climate, 2025, L1_02). Solar's share of national renewable additions—approximately 25% in the Reference scenario and 30% in the Accelerated scenario—yields meaningful investment implications. Based on national renewable investment expectations of $14–$20 billion annually (2025–2035), solar's portion equates to $3.5 billion/year (Reference) and $6.0 billion/year (Accelerated), with cumulative investments reaching $36 billion (Reference) and $61.5 billion (Accelerated) by 2035 (CanREA & Dunsky Energy + Climate, 2025, L1_02).

Employment impacts follow similar proportional allocations. National renewable additions generate 250,000–350,000 job-years over the decade, of which 62,500 (Reference) and 105,000 (Accelerated) are attributable to solar. Derived employment intensity of 3.3–3.75 job-years per MW aligns with global solar deployment norms and provides a defensible metric for economic development analysis (CanREA & Dunsky Energy + Climate, 2025, L1_02). See Figure 10 and 11 below.

Figure 10 *Figure 11*

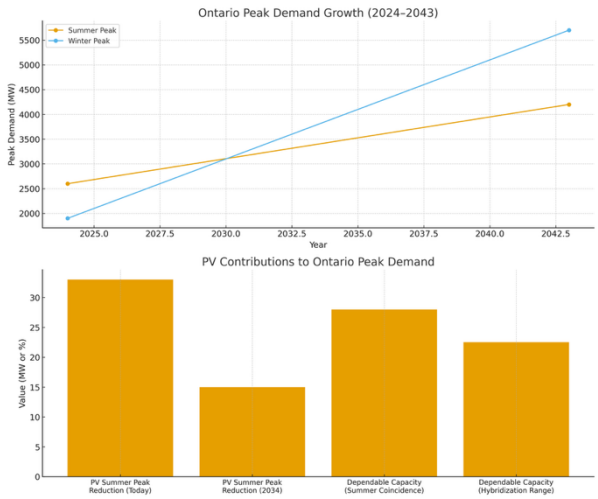


These investment and labour-market implications demonstrate that PV deployment contributes not only to Ontario's climate and reliability objectives but also strengthens economic competitiveness and workforce transitions aligned with provincial electrification ambitions.

3.3 PV Economics in the Context of Electrification and Load Growth

Ontario's evolving load profile shapes the economic value of PV in ways not captured by cost curves alone. The GTA North IRRP projects dramatic increases in both summer and winter peak demand: summer peaks rise from 2,600 MW (2024) to 4,200 MW (2043), while winter peaks—driven by building electrification—surge from 1,900 MW to 5,700 MW over the same period (IESO, 2025, L2_03). Winter becomes the dominant constraint after 2035, fundamentally altering the seasonal value proposition of PV. See Figure 12 *"Ontario Peak Demand Growth (2024-2043)"*.

Figure 12



At present, distributed PV reduces regional summer peaks by 33 MW, but this declines to 14–15 MW by 2034 as legacy FIT and microFIT contracts expire. This represents less than 1% of regional extreme-weather peak, underscoring that PV's direct load-modifying impact is modest relative to system-wide electrification trends (IESO, 2025, L2_03). However, when paired with energy storage and demand-side measures, PV-generated capacity value becomes more material within the IRRP's hybrid planning framework. This aligns with national and NREL hybridization findings that standalone PV typically provides between 15 and 30% dependable capacity depending on tilt, orientation, and siting. See Figure 12 *"PV Contributions to Ontario Peak Demand (2024-2043)"*.

The IESO's Distributed Energy Resources Potential Study assigns PV a 28% summer peak-coincidence factor, meaning that on average roughly 28% of nameplate PV output can be counted on during coincident summer peak hours (Dunsky Energy Consulting, 2022, L1_04). Complementary hybridization analysis finds that standalone PV can provide a dependable-capacity contribution in the range of 15–30%, depending on tilt, orientation, and siting (National Renewable Energy Laboratory, 2024, L1_06). In simpler terms, this means that well-sited PV can reliably shoulder roughly a quarter of coincident summer peak demand and can be credited with 15–30% of its nameplate capacity during the specific summer windows when Ontario's system is most stressed—making it a technically credible contributor to distribution-level

deferral strategies when integrated into broader DER portfolios. See Figure 12 *"PV Contributions to Ontario Peak Demand (2024-2043)"*.

3.4 Cost-Effectiveness of DER Portfolios and Implications for PV

The strongest empirical evidence for the economic value of DER-based deferral in Ontario comes from the York Region NWA Demonstration. Though the pilot was dominated by demand response and thermal DERs—not PV—it provides critical cost benchmarks for local capacity provision relevant to future solar-plus-storage configurations (IESO & ICF, 2024, L2_01).

Key performance and pricing outcomes include 15 activation events, 366 MWh delivered, 83% availability, 85% performance accuracy, Local Capacity Auction clearing prices dropping from $0.64/kW-day (Year 1) to $0.40/kW-day (Year 2), an equivalent DER procurement cost of $100,000/MW-year, and net system benefits ranging from –$49,000 to +$382,000/MW-year (2027) and up to +$515,000/MW-year (2032) (IESO & ICF, 2024, L2_01). These values reflect avoided generation capacity, transmission deferral, and distribution deferral, directly aligning with the OEB FEI's multi-layer BCA categories (OEB, 2023, L2_02).

The implication for PV is clear: while standalone PV offers limited capacity value due to intermittency, PV paired with storage—as costs continue to decline—can be compared against these DER cost benchmarks. Given utility-scale LCOE trajectories approaching 20 USD/MWh (NREL, 2024, L1_01) and Ontario-specific LCOE trending toward 33–50 CAD/MWh (CanREA & Dunsky Energy + Climate, 2025, L1_02), solar will increasingly compete with, or complement, DER portfolios providing local capacity value. See Figure 13 below.

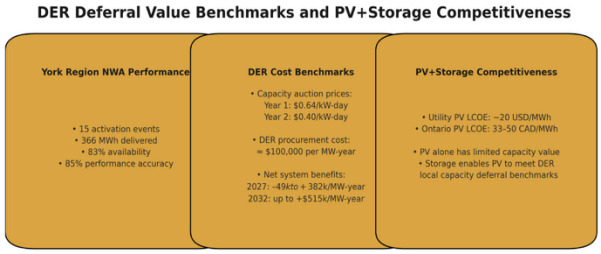


Figure 13

3.5 Regulatory Alignment and Economic Integration

The OEB's Framework for Energy Innovation and the IESO's NWA Screening Tool formalize solar and storage as economically evaluable resources within Distribution System Plans and IRRPs. Key principles include "costs should follow benefits," standardized multi-test Benefit-Cost Analysis, permitted use of discounted cash-flow methods until final BCA guidance is issued, and allowance for utility incentives (shared savings, scorecard mechanisms, margin-on-payments) to mitigate disincentives for non-capital DER adoption (OEB, 2023, L2_02).

The GTA North IRRP applies this regulatory framework by identifying 60–80 MW of deferrable capacity across key subregions and explicitly incorporating DG and DSM into long-term planning (IESO, 2025, L2_03). Together, these regulatory and economic factors position PV as a maturing, increasingly competitive resource within Ontario's evolving grid—both as a bulk energy contributor and as part of DER-integrated local capacity strategies.

4. Economic Characteristics and Cost Data

This section synthesizes the quantitative evidence on DER-based non-wires alternatives (NWAs) from Ontario's recent pilots and planning documents and translates it into an economic frame for rooftop PV and PV-plus-storage portfolios in York Region. Rather than constructing a bottom-up levelized cost of energy (LCOE) model for PV-plus-storage, the analysis treats the York Region NWA Demonstration as a revealed-preference benchmark for local capacity value and the Framework for Energy Innovation (FEI) and GTA North IRRP as the regulatory and planning context that governs how those values would actually be realized (Independent Electricity System Operator [IESO] & ICF, 2024, L2_01; Ontario Energy Board [OEB], 2023, L2_02; IESO & Technical Working Group, 2025, L2_03).

4.1 Capital and Operating Cost Analogues from the York Region NWA Demonstration

In the York Region NWA Demonstration, Alectra and the IESO procured 10 MW of local capacity in 2021 and 15 MW in 2022 through Local Capacity Auctions that were explicitly designed as substitutes for a targeted distribution upgrade (IESO & ICF, 2024, L2_01, pp. 6, 20–21). Across the two years:

- The auctions were oversubscribed, with 19 MW of bids for a 10 MW target in Year 1 and 22.8 MW for a 15 MW target in Year 2 (IESO & ICF, 2024, L2_01, p. 6).
- Clearing prices were $0.64/kW-day in the first year and $0.40/kW-day in the second (IESO & ICF, 2024, L2_01, p. 21). Normalized over a six-month commitment period and adjusted for actual availability, this translated to approximately $80,000/MW and $50,000/MW per commitment season, respectively.

Availability payments, adapted from the IESO's capacity-auction design, accounted for roughly 71% of all settlement payments, providing participants with a stable, capacity-like revenue stream (IESO & ICF, 2024, L2_01, pp. 36–37). Variable, activation-based Local Energy payments (with a ceiling of $2.00/kWh and average accepted offers of $1.81/kWh in 2021 and $1.19/kWh in 2022) represented the operating or dispatch-linked component of compensation (IESO & ICF, 2024, L2_01, pp. 22–23, 36–37).

From an economic-characterization perspective, this split maps naturally onto a PV-plus-storage portfolio:

- **Capacity-like revenue:** availability payments (in $/MW-season or $/MW-year) are the closest analogue to recovering fixed capital and fixed operating expenses for a PV-plus-storage NWA portfolio that commits dependable capacity to the local system.
- **Energy- and performance-linked revenue:** Local Energy payments at a high local scarcity price represent the operating-margin side of the portfolio—compensation for actual dispatch during constrained hours and for meeting performance obligations.

In the York Region pilot, portfolio-level availability of 83% and performance accuracy of 85% were used as de-rating factors in the cost-benefit analysis (IESO & ICF, 2024, L2_01, pp. 7–8, 25, 29). Any rooftop PV-plus-storage portfolio would have to meet comparable availability and performance metrics to access similar capacity and energy revenues. The numbers therefore provide empirical bounds on the annual revenue per MW that a PV-plus-storage NWA could plausibly target in the same geography and regulatory context, even though the demonstration's actual mix was dominated by demand response and small thermal resources (IESO & ICF, 2024, L2_01, p. 6).

4.2 System Value and Avoided-Cost Components

The York Region NWA Demonstration also provides an integrated benefit-cost analysis (BCA) of DER portfolios as alternatives to conventional "wires" investments. ICF's evaluation, prepared for the IESO, models the net economic effect of DERs under multiple load-growth scenarios and planning horizons (IESO & ICF, 2024, L2_01, pp. 8–9).

For the 2027 horizon, the report presents net economic value in $/MW-year terms (all avoided costs minus DER procurement costs):

- **Slow Growth 2027:** –$49,000/MW-year (DERs not economic under low-growth, low-constraint conditions).
- **Base Case 2027:** +$191,000/MW-year, indicating a clearly positive net benefit.
- **High Growth 2027:** +$382,000/MW-year, showing increasing value as constraints bind more often (IESO & ICF, 2024, L2_01, p. 8).

By 2032, net benefits increase to +$227,000/MW-year (Base Case) and +$515,000/MW-year (High Growth), reflecting rising avoided-capacity and deferral values over time (IESO & ICF, 2024, L2_01, p. 9).

The underlying value stack in the 2027 Base Case is decomposed as:

- Avoided generation capacity: $192,000/MW-year
- Transmission deferral: $98,000/MW-year
- Distribution deferral: $67,000/MW-year
- Avoided energy and losses: about $720/MW-year
- Less DER procurement cost: –$167,000/MW-year

for a net of approximately +$191,000/MW-year (IESO & ICF, 2024, L2_01, p. 8). In other words, 85–90% of the total benefit is driven by capacity-related deferral (generation, transmission, and distribution), while avoided energy and loss reductions are negligible in comparison (IESO & ICF, 2024, L2_01, pp. 8–9).

These results have three important implications for a rooftop PV-plus-storage NWA portfolio:

1. **Capacity, not energy, is the primary value driver.** The York Region BCA confirms that capacity deferral dominates the economic case; energy value is less than 1% of total benefit. Any PV-plus-storage NWA design should therefore be optimized for dependable peak-period capacity contribution (e.g., late summer afternoons and evenings), not for bulk energy arbitrage (IESO & ICF, 2024, L2_01, pp. 8–9).
2. **Local DER procurement can be cost-competitive with bulk capacity.** After normalizing the local capacity-auction prices to an annual basis and accounting for the six-month term and availability de-rating, ICF concludes that effective local cost is on the order of $100,000/MW-year, which is "within 10–15%" of IESO's provincial capacity-auction costs for comparable periods (IESO & ICF, 2024, L2_01, pp. 8, 21). When stacked against net benefits of $191,000–$382,000/MW-year in 2027 and up to $515,000/MW-year by 2032, this implies a benefit-cost ratio greater than 2:1 in the Base Case and significantly higher under high-growth conditions (IESO & ICF, 2024, L2_01, pp. 8–9).
3. **Portfolio composition is flexible as long as performance is delivered.** The demonstration's actual resource mix was dominated by demand response (≈80–90% of procured MW), with the

remainder provided by small thermal generation and no storage (IESO & ICF, 2024, L2_01, p. 6). However, the BCA methodology is resource-agnostic: benefits are calculated at the portfolio level based on delivered, de-rated capacity and energy. A future PV-plus-storage portfolio that meets similar availability, performance, and dispatch requirements would plug into the same $/MW-year valuation framework, subject to its own cost structure and interconnection constraints.

Taken together, the York Region numbers demonstrate that DER-based NWAs can be not just technically viable but economically attractive, provided they are targeted to high-growth, capacity-constrained locations and are procured through competitive mechanisms that produce prices near provincial capacity benchmarks (IESO & ICF, 2024, L2_01).

The York Region NWA value-stack waterfalls below (Figure 14 and 15) ties sections 4.1 and 4.2 together nicely and provides a direct, visual readout of what drives DER economics in Ontario. The 2027 Base Case chart shows that almost the entire net benefit comes from avoided generation, transmission, and distribution capacity, with avoided energy playing a negligible role. The 2032 Base Case and 2032 High Growth waterfalls extend this logic forward in time: as load constraints intensify, each avoided-capacity category becomes larger, while DER procurement costs rise more modestly. This produces sharply higher net benefits—$227k/MW-year in 2032 Base Case and $515k/MW-year in High Growth—demonstrating that DER portfolios become increasingly valuable in capacity-constrained futures. Read together, the waterfalls illustrate the central insight of this section: PV-plus-storage NWAs are economically meaningful only to the extent that they deliver dependable peak-period capacity, because capacity—not energy—is where 85–90% of total system value is derived.

Figure 14

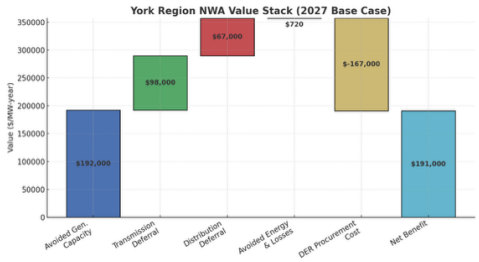
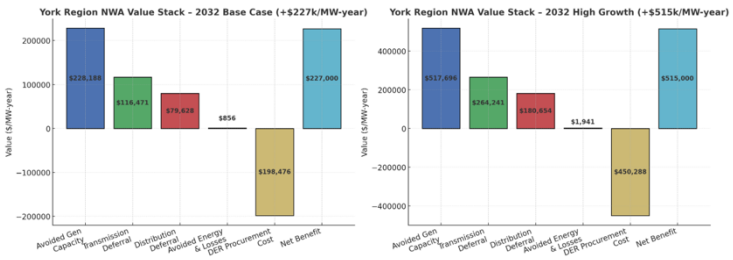


Figure 15



4.3 Alignment with the OEB's BCA Framework and Utility Incentives

The Ontario Energy Board's Framework for Energy Innovation turns the type of project-specific BCA used in the York Region NWA pilot into a province-wide regulatory methodology (OEB, 2023, L2_02). In section 5.1, the OEB explicitly states that the BCA Framework is intended to "support consistent evaluation of DER solutions across distributors and reduce uncertainty about how DER proposals will be assessed … in rate applications" (OEB, 2023, L2_02, p. 18).

The FEI BCA framework is built on two core design elements that are directly relevant to PV-plus-storage NWAs:

- **Cost-allocation principle – "Costs should follow benefits."** FEI confirms that the primary test for cost recovery is the Distribution Service Test, which evaluates costs and benefits from the perspective of the implementing distributor's customers (OEB, 2023, L2_02, pp. 19–21). Broader system impacts—transmission, resource adequacy, and general energy system benefits—can be

quantified in a broader Energy System Test, but recovery through distribution rates should reflect benefits accruing to the distributor's own customers (OEB, 2023, L2_02, pp. 19–21, 39). This is conceptually identical to the way ICF's York Region analysis stacks generation, transmission, and distribution deferral value and then tests whether net benefits are positive on a $/MW-year basis (IESO & ICF, 2024, L2_01, pp. 8–9).

- **Multi-layer impact scope.** FEI states that the BCA Framework "will ultimately allow for consideration of all the categories of system impacts (distribution service, DER host, transmission service, resource, and general energy system impacts)" (OEB, 2023, L2_02, pp. 20–21). The York Region NWA Demonstration already anticipates this structure by disaggregating avoided generation, transmission, and distribution deferral components and then aggregating them into a single net benefit figure per MW (IESO & ICF, 2024, L2_01, pp. 8–9).

For a rooftop PV-plus-storage NWA, this alignment means that:

- The same avoided-cost components used in York Region—generation capacity, transmission and distribution deferral, and marginal energy—can be carried into a formal FEI-compliant BCA, provided that the PV-plus-storage portfolio can demonstrate comparable reliability and performance.
- The "costs follow benefits" principle gives a clear boundary for rate recovery: if a PV-plus-storage portfolio is procured to address a specific York Region constraint, its costs can, in principle, be recovered from Alectra's customers to the extent that the BCA shows positive net value for that customer group (OEB, 2023, L2_02, pp. 19–21).

FEI's utility incentives workstream further acknowledges the structural barrier that traditional cost-of-service regulation can create for third-party DER solutions. The report notes that DERs will "often, from a utility point of view, be non-capital in nature," meaning distributors may lose a traditional rate-base return when DERs displace capital projects (OEB, 2023, L2_02, p. 24). To address this, FEI:

- Encourages distributors to use deferral accounts to track DER-related planning and procurement costs before rebasing, provided they meet causation, materiality, and prudence criteria (OEB, 2023, L2_02, pp. 27–28).
- Identifies and endorses testing of three incentive mechanisms within current rate-setting rules: shared savings, performance-target/scorecard-based incentives, and margin on payments (OEB, 2023, L2_02, pp. 28–29).

These instruments are particularly relevant for PV-plus-storage NWAs. A distributor that procures a rooftop PV-plus-storage portfolio to replicate York Region-style net benefits (e.g., +$191,000 to +$382,000/MW-year in 2027 scenarios) could, in principle, recover its procurement and integration costs and potentially earn a performance-based incentive without needing to own the physical assets (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02). The result is a regulatory pathway where PV-plus-storage NWAs are evaluated using a standardized BCA and utilities are not financially penalized for choosing the non-wires option.

4.4 Planning Context and Scale of PV-Plus-Storage NWAs in GTA North

The GTA North (York Region) Integrated Regional Resource Plan provides the long-term planning backdrop into which any PV-plus-storage NWA must fit. The IRRP forecasts that regional net extreme-weather coincident peaks will grow from roughly 2,600 MW (summer) and 1,900 MW (winter) in 2024 to about 4,200 MW (summer) and 5,700 MW (winter) by 2043 (IESO & Technical Working Group, 2025, L2_03, Fig. 7, pp. 22–28). Over the planning horizon, winter peaks become the dominant constraint

due to electrification of heating, even though summer remains the primary driver of investment timing through the early 2030s (IESO & Technical Working Group, 2025, L2_03).

Distributed generation—primarily legacy FIT and microFIT solar PV—provides only a modest reduction in regional peaks. Figure 6 of the IRRP shows summer DG peak-reduction falling from about 33 MW in 2024 to around 14 MW by the mid-2030s, with winter reductions declining from about 17 MW to roughly 13 MW as contracts expire (IESO & Technical Working Group, 2025, L2_03, pp. 26–27). In percentage terms, this is less than 1–2% of regional peak load. Combined with incremental demand-side management (eDSM), the IRRP estimates 35–40 MW of peak-reduction potential by the early 2030s—again, less than 2% of total demand but still materially relevant for local station-level constraints (IESO & Technical Working Group, 2025, L2_03, pp. 22–28, 53–74).

On the supply side, the IRRP projects region-wide growth of roughly 2,100 MW (summer) and 3,400 MW (winter) over the 20-year horizon, of which about 1,500 MW (summer) and 2,200 MW (winter) are already addressed through committed station and transmission projects (IESO & Technical Working Group, 2025, L2_03, pp. 33–51). The remaining ≈600 MW of summer growth and ≈1,200 MW of winter growth represent the "residual" need that could, in principle, trigger further wires reinforcements or be partially addressed by NWAs, depending on how actual load materializes (IESO & Technical Working Group, 2025, L2_03, pp. 33–51).

The IRRP applies the IESO Non-Wires Alternatives Screening Tool to sub-regional needs using three criteria: magnitude of deferrable cost, load-duration characteristics of the constraint, and timing and lead time (IESO & Technical Working Group, 2025, L2_03, pp. 53–74). These criteria closely mirror the IESO's province-wide *Guide to Assessing Non-Wires Alternatives*, which formalizes a standardized NWA screening process and ensures that regional plans assess DER portfolios against traditional wires solutions on an equivalent-reliability and cost basis (Independent Electricity System Operator [IESO], 2023, L1_07). On this basis, three sub-regions—Northern York, Markham, and Parkway–Buttonville, are "screened in" for NWA consideration (IESO & Technical Working Group, 2025, L2_03, Tables 7–8). For these areas, the IRRP concludes that:

- Large, transmission-connected NWAs are not currently viable due to existing transmission congestion and connection limits.
- Incremental eDSM and DER portfolios can provide localized capacity relief equivalent to roughly 60–80 MW (rising toward ~100 MW by the mid-2030s) across the screened-in sub-regions, potentially deferring or downsizing some station investments (IESO & Technical Working Group, 2025, L2_03, pp. 59–77, Table 10).

The Kleinburg–Kirby 230 kV double-circuit transmission link and the upgrade of Kleinburg TS to a 500/230 kV autotransformer station, estimated at approximately $400 million, remain essential bulk reinforcements (IESO & Technical Working Group, 2025, L2_03, pp. 59–77). The IRRP therefore explicitly adopts a hybrid planning approach: transmission expansion plus targeted DER/eDSM portfolios to manage local peaks and defer subsequent infrastructure where feasible (IESO & Technical Working Group, 2025, L2_03, pp. 59–77, 75–76).

For a rooftop PV-plus-storage NWA, the implications are straightforward:

- The scale of opportunity is on the order of tens of megawatts, not hundreds, of local deferrable capacity in the 2030–2035 window—precisely the range where the York Region NWA Demonstration's 10–15 MW procurements already provided empirical cost and performance data (IESO & ICF, 2024, L2_01; IESO & Technical Working Group, 2025, L2_03).

- Because DER and eDSM contributions are already embedded in the IRRP's net extreme-weather forecast, a PV-plus-storage NWA must be evaluated as an incremental resource—one that can shift the timing or scale of specific station projects, but not replace major bulk system investments (IESO & Technical Working Group, 2025, L2_03).
- Any PV-plus-storage NWA portfolio would also need to satisfy the DER Connection Procedures (Preliminary Consultation, CIA, CCA, and Build & Energization), meaning that its cost structure must incorporate the same interconnection, protection, and capacity-reservation steps that are already standardized for other DERs (OEB, 2025, L1_08).

Overall, the combination of York Region NWA Demonstration economics, the FEI BCA and incentives framework, and the GTA North IRRP planning context shows that rooftop PV-plus-storage NWAs in York Region would be evaluated against a clear benchmark:

- Local DER capacity can be procured at roughly $100,000/MW-year and still deliver net benefits of $191,000–$382,000/MW-year in 2027, rising to $227,000–$515,000/MW-year by 2032 under plausible growth scenarios (IESO & ICF, 2024, L2_01).
- The regulatory machinery now exists to test such portfolios under a standardized BCA, align cost recovery with beneficiaries, and offer utilities transitional incentives for choosing non-wires options (OEB, 2023, L2_02).
- The regional planning process has begun to systematically screen for and scale such options in high-growth nodes like Northern York, Markham, and Parkway–Buttonville (IESO & Technical Working Group, 2025, L2_03).

Within this framework, rooftop PV-plus-storage is best understood not as a stand-alone alternative to major transmission projects, but as a targeted, capacity-oriented NWA building block that can compete economically with other DERs when designed to meet the same availability and performance obligations demonstrated in York Region.

---

5. Regulatory and Policy Landscape

Ontario's regulatory environment for evaluating solar photovoltaics (PV) and distributed energy resources (DERs) as non-wires alternatives is undergoing a foundational transition, driven by the Ontario Energy Board's Framework for Energy Innovation, the IESO's DER Market Design and operational pilots, and the integration of DERs into regional planning through the Integrated Regional Resource Planning process. These policy instruments collectively shape how PV and related resources are assessed, valued, and procured within the province. The following subsections integrate evidence from the FEI BCA Framework (OEB, 2023, L2_02), utility incentives guidelines (OEB, 2023, L2_02), and the GTA North IRRP (IESO & Technical Working Group, 2025, L2_03), alongside operational proof from the York Region NWA Demonstration (IESO & ICF, 2024, L2_01).

5.1 Benefit-Cost Analysis and Valuation Frameworks

The OEB's Framework for Energy Innovation marks Ontario's first formal attempt to standardize how DERs, including solar PV, are evaluated as non-wires alternatives in regulated distribution planning. The OEB explicitly states that the purpose of the BCA Framework is to "support consistent evaluation of DER solutions across distributors and reduce uncertainty about how DER proposals will be assessed … in rate applications" (OEB, 2023, L2_02, p. 18).

The FEI grounds its valuation in the regulatory principle that "costs should follow benefits," meaning that DER project costs should be borne primarily by the customers who receive the benefits (OEB, 2023, L2_02, pp. 19–21). While the Framework allows consideration of transmission, resource adequacy, and general energy system benefits, the OEB clarifies that the distributor-customer perspective remains the primary determinant for rate approval.

The BCA Framework adopts a multi-test structure, beginning with a Distribution Service Test—focused on local system impacts—and extending to an Energy System Test encompassing broader avoided costs (OEB, 2023, L2_02, p. 22). Each test includes standardized components: purpose and use, information requirements, cost-effectiveness test, standardized methods, and reporting requirements (OEB, 2023, L2_02, Fig. 7). Phase 1 guidance is to be applied by all distributors by the end of FY 2023/24, with expanded energy-system valuation methods anticipated during FY 2024/25 (OEB, 2023, L2_02, pp. 21–23).

Importantly, the FEI emphasizes that distributors should not wait for final guidance. Pending completion of the full BCA Framework, utilities may use a conventional discounted cash-flow analysis, provided they transparently describe how benefits and costs were considered (OEB, 2023, L2_02, p. 23).

Together, these policies create the economic and methodological foundation that will govern how PV, storage, and hybrid DER portfolios are assessed in NWA screening and regional planning.

5.2 Utility Remuneration, Incentives, and Cost Recovery

A major barrier to DER uptake in regulated systems is the misalignment between utility incentives and non-capital solutions. The FEI explicitly identifies the issue: utilities traditionally "earn profits by building assets," whereas DERs "will often … be non-capital in nature" (OEB, 2023, L2_02, p. 24). Without reform, utilities are financially disincentivized from selecting DERs—even when they are more cost-effective for customers.

The FEI responds through two coordinated mechanisms:

- **Cost recovery for DER-related OM&A and planning activities.** The OEB clarifies that DER integration is becoming a routine utility function. Accordingly, planning, procurement, and administrative costs "will generally be treated the same as other capital and OM&A spending" (OEB, 2023, L2_02, pp. 27–28). This effectively normalizes DER integration within standard utility operations, reducing uncertainty about how DER-related expenses will be treated during rate applications. In the interim, distributors may apply for deferral accounts to record material DER-related costs, subject to causation, materiality, and prudence criteria.
- **Testing incentive mechanisms for third-party DERs.** The OEB selects three near-term incentive mechanisms for pilot testing: shared savings, performance-target/scorecard-based incentives, and margin on DER payments (OEB, 2023, L2_02, pp. 28–29). These mechanisms operate within the existing rate-setting framework and will be evaluated over 24–36 months, serving as an incremental step toward broader remuneration reform aligned with electrification and energy transition priorities (OEB, 2023, L2_02, p. 29).

For PV specifically, these incentives establish the conditions under which utilities may financially benefit from procuring third-party solar capacity as part of a broader NWA or system-planning strategy.

5.3 Transmission–Distribution Coordination and Local Market Design

Ontario's regulatory direction is also shaped by empirical findings from the York Region NWA Demonstration, which remains the province's most advanced test of distribution-level DER markets and coordination processes. Under the Total-DSO model, Alectra acted as a distribution system operator, administering Local Capacity, Local Energy, and Local Reserve Auctions (IESO & ICF, 2024, L2_01, pp. 17–18). The demonstration used a simulated environment to test interoperability between Alectra and the IESO, confirming that T–D coordination is technically feasible using existing operational frameworks (IESO & ICF, 2024, L2_01, pp. 22–23).

Key outcomes influencing the regulatory landscape include competitive local auctions clearing 10 MW (2021) and 15 MW (2022), oversubscription rates of 1.9× and 1.5×, availability of 83%, performance accuracy of 85%, and Local Energy Prices averaging $1.81/kWh and $1.19/kWh—far above wholesale LMPs (IESO & ICF, 2024, L2_01, pp. 6, 22–23, 29). These findings supply the IESO and OEB with empirical data demonstrating that DERs can reliably provide peak-support services, local auctions can reveal locational value, and distribution-level markets can operate under coordinated DSO-administered frameworks (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02).

5.4 Alignment with Regional Planning and Long-Term Resource Adequacy

The GTA North IRRP integrates DERs and non-wires alternatives directly into the IESO's regional planning process. For the first time, the IRRP applies the IESO's formal Non-Wires Alternatives Screening Tool, evaluating needs based on deferrable cost magnitude, load-duration profile, and timing (IESO & Technical Working Group, 2025, L2_03, pp. 53–74). Using this methodology, three subregions—Northern York, Markham, and Parkway–Buttonville—were "screened in" for further NWA assessment.

The IRRP identifies combined summer heat-wave peaks rising from ~2,600 MW (2024) to ~4,200 MW (2043) and winter peaks rising from ~1,900 MW to ~5,700 MW, with DER/DG contributions declining as legacy contracts expire (IESO & Technical Working Group, 2025, L2_03, Fig. 6–7). Although large-scale DERs cannot replace major transmission projects due to congestion at Claireville TS and 230 kV corridor limits, the IRRP determines that incremental DER and eDSM portfolios can defer 60–80 MW of station-level upgrades across the screened-in subregions and identifies a long-term residual capacity need of approximately 600 MW (summer) and 1,200 MW (winter) even after committed reinforcements (IESO & Technical Working Group, 2025, L2_03, Sec. 6.4, Tables 7–10).

Together, the FEI, NWA Demonstration, and GTA North IRRP establish a coherent policy architecture that directly shapes how solar PV is assessed within Ontario's evolving grid planning and regulatory environment.

---

6. Adoption and Market Potential

Ontario's adoption potential for solar PV and distributed energy resources is shaped by evolving system economics, declining technology costs, and the province's formal integration of DERs into regulatory, market, and planning frameworks. Across the Level 1 and Level 2 sources, three consistent themes emerge:

- PV economics are improving more rapidly than legacy assumptions.
- DER operational performance is empirically validated through the York Region NWA Demonstration.
- Regional electrification pressures are creating new capacity needs that DERs can partially address.

Together, these factors point to a growing, though uneven, adoption landscape for PV and DER resources in Ontario.

6.1 Economic Competitiveness and Cost Trajectories

The NREL ATB 2024 provides the most current cost trajectories for solar PV and remains central to evaluating adoption potential. Under the Class 5 Moderate (Mid) scenario—chosen for consistency with IESO and NRCan planning assumptions—PV capital costs fall substantially across utility-scale, commercial, and residential segments (National Renewable Energy Laboratory [NREL], 2024, L1_01).

For utility-scale PV, capital costs decline from $1,611/kW in 2023 to $895/kW by 2035, with fixed O&M dropping to $15/kW-year and LCOE reaching $26/MWh (NREL, 2024, L1_01). By 2050, utility-scale costs fall even further to $683/kW with LCOE at $20/MWh, confirming long-term competitiveness relative to Ontario's typical wholesale supply costs. Commercial and residential systems follow similar but higher-cost trajectories, with residential systems declining from $2,682/kW to $1,703/kW (2035) and $1,208/kW (2050) (NREL, 2024, L1_01).

Contextualized within Canada's broader energy landscape, Smart Prosperity Institute notes that solar PV already falls below the cost of gas-fired generation by 2026, becoming cost-competitive nationally (Smart Prosperity Institute, 2021, L1_03). The Canada Energy Regulator's Evolving Scenario further projects that national PV capacity will grow from 3 GW in 2020 to 8 GW by 2030 and 20 GW by 2050, representing a sixfold increase (Smart Prosperity Institute, 2021, L1_03).

These trends establish a clear economic foundation: solar PV is increasingly financially viable at scale, and long-term projections show sustained growth driven by falling costs and expanding policy support. Natural Resources Canada's building-stock assessment reinforces this trend, concluding that Canada's existing building rooftops represent a substantial untapped technical resource for PV deployment, with significant potential concentrated in provinces with large commercial and residential building stocks such as Ontario (Gaucher-Loksts & Pelland, 2024, L1_05).

6.2 Operational Feasibility and DER Performance

Ontario's most important empirical evidence for DER viability comes from the York Region NWA Demonstration, which tested competitive DER procurement under a coordinated IESO–DSO operational model (IESO & ICF, 2024, L2_01). The demonstration provides adoption-relevant insights into DER reliability, participant experience, market depth, and price discovery.

Key operational outcomes include 15 DER activations over two years, 366 MWh delivered, 83% average availability, 85% performance accuracy, and oversubscription of local capacity auctions (19 MW vs. 10 MW target in 2021; 22.8 MW vs. 15 MW target in 2022) (IESO & ICF, 2024, L2_01, Table 1, pp. 7–8, 29). This oversubscription demonstrates clear market appetite and meaningful DER supply within a single distribution region.

Local energy auctions revealed strong locational value—with accepted offers averaging $1.81/kWh (2021) and $1.19/kWh (2022), far above typical wholesale LMPs of $0.10–$0.15/kWh (IESO & ICF, 2024, L2_01). While these prices are not intended to be future DLMP benchmarks, they demonstrate that DERs can provide high economic value during local constraints. Participant feedback shows strong readiness for broader DER adoption, with key barriers identified as telemetry and metering costs, short contract durations, baseline and M&V complexity, and limited data visibility—primarily procedural rather than technological barriers (IESO & ICF, 2024, L2_01, pp. 24–25, 65).

Collectively, the demonstration verifies that DERs—including demand response, CHP, and distributed storage—can function as reliable capacity resources and attract sustained participation under supportive market designs (IESO & ICF, 2024, L2_01).

6.3 Distribution System Constraints and Regional Adoption Potential

The GTA North IRRP provides detailed regional planning evidence showing where and how DERs—including solar PV—can play a meaningful role in relieving system stress (IESO & Technical Working Group, 2025, L2_03). Ontario's second-fastest-growing load pocket, the GTA North region, forecasts summer peaks rising from ~2,600 MW (2024) to ~4,200 MW (2043) and winter peaks rising from ~1,900 MW to ~5,700 MW over the same period (IESO & Technical Working Group, 2025, L2_03, Fig. 7). This seasonal inversion—where winter becomes the dominant constraint due to heating electrification—is central to evaluating future solar adoption because winter PV output is lower.

Nevertheless, the IRRP identifies measurable, though modest, load-modifying contributions from legacy distributed generation (primarily FIT/microFIT solar). DG reduces summer peaks from 33 MW (2024) to 14 MW (2034+) and winter peaks from 17 MW to 13 MW (IESO & Technical Working Group, 2025, L2_03, Fig. 6). While these contributions represent ≤1% of regional peak and decline over time, their explicit inclusion in the IRRP's net-planning forecast demonstrates that distributed PV is already an operational planning input. Because these DG contributions are already embedded in the IRRP's net-load forecast, any new PV or DER adoption must be evaluated as incremental to the existing baseline.

At the provincial scale, the DER Potential Study estimates 122 GW of rooftop technical PV potential and 56 GW of quasi-economic potential in Ontario, confirming that the physical and economic headroom for additional PV deployment is orders of magnitude larger than the modest DG volumes currently embedded in regional plans (Dunsky Energy Consulting, 2022, L1_04).

More importantly, the IRRP applies the IESO's Non-Wires Alternatives Screening Tool, screening in Northern York, Markham, and Parkway–Buttonville for NWA evaluation (IESO & Technical Working Group, 2025, L2_03, Tables 7–8). The IRRP concludes that 60–80 MW of incremental eDSM and DER capacity could defer or downsize near-term station upgrades, while large-scale DER projects remain constrained by transmission limits; DERs are therefore best suited as localized capacity resources rather than bulk replacements (IESO & Technical Working Group, 2025, L2_03, Sec. 7.5). The plan also identifies a long-term unserved residual need of ~600 MW (summer) and ~1,200 MW (winter) even after committed wires reinforcements (IESO & Technical Working Group, 2025, L2_03, Sec. 6.4).

6.4 Long-Term Market Growth and Adoption Outlook

Combined across all sources, Ontario's adoption potential reflects multiple layers of evidence:

- PV technology is rapidly declining in cost (NREL, 2024, L1_01; Canadian Renewable Energy Association [CanREA] & Dunsky Energy + Climate, 2025, L1_02).

- DERs have demonstrated reliable operational performance in Ontario (IESO & ICF, 2024, L2_01).
- Formal regulatory frameworks now support DER deployment (OEB, 2023, L2_02).
- Regional forecasts identify growing capacity needs that DERs can meaningfully address (IESO & Technical Working Group, 2025, L2_03).
- National projections show sustained long-run solar growth (Smart Prosperity Institute, 2021, L1_03).

Taken together, these findings depict a province with strong economic conditions for PV adoption, proof-of-concept DER reliability, maturing regulatory structures, and rapidly growing system needs. While PV alone will not meet Ontario's winter peaks, the integrated evidence indicates substantial room for PV-centric DER portfolios to support non-wires alternatives, reduce local constraints, and contribute meaningfully to long-run supply diversity.

---

7. Comparative Analysis and Synthesis

Ontario's long-term system needs, emerging market structures, and evolving technology costs create a complex but increasingly coherent picture of how solar PV and distributed energy resources can contribute to future grid capacity. A comparative synthesis of the Level 1 (PV cost and market outlook) and Level 2 (DER/NWA performance, regulatory frameworks, and regional planning) sources reveals several converging insights regarding economic viability, operational feasibility, and system integration potential.

7.1 Comparative Economic Trajectories Across Sources

Across multiple evidence streams, solar PV exhibits sustained and substantial cost declines. NREL ATB 2024 presents a consistent downward trajectory across utility-scale, commercial, and residential PV, with utility-scale capital costs falling from $1,611/kW (2023) to $895/kW (2035) and LCOE reaching $26/MWh, and further to $683/kW by 2050 (NREL, 2024, L1_01). Smart Prosperity Institute confirms that PV had already reached cost parity with gas-fired generation by 2026 and is expected to continue improving in competitiveness through 2050 (Smart Prosperity Institute, 2021, L1_03).

CanREA 2025 reinforces these trends with a Canadian, policy-specific perspective, showing Reference-scenario CapEx declines of 7–9% by 2030–2035, Accelerated-scenario declines of 11–13% over the same period, and LCOE estimates falling to $49–52/MWh (2030–2035) (CanREA & Dunsky Energy + Climate, 2025, L1_02). Collectively, these results indicate clear economic momentum behind PV deployment and establish cost-based justification for integration into Ontario's long-term planning.

Comparatively, DER procurement costs demonstrated through the York Region NWA Demonstration align with broader trends in cost-effectiveness. Local capacity auctions cleared at normalized values of roughly $100,000/MW-year when adjusted for availability and contract length—within 10–15% of IESO's wholesale capacity auction prices (IESO & ICF, 2024, L2_01). These values demonstrate that DER-based solutions are economically comparable to traditional bulk capacity procurement.

7.2 Operational Performance and Integration Feasibility Across Evidence

The operational viability of DERs is best demonstrated in the York Region NWA Demonstration, which provides Ontario's first real-world performance dataset. As summarized earlier, DER portfolios achieved 83% availability, 85% performance accuracy, delivered 366 MWh, and attracted oversubscribed capacity bids in both years (IESO & ICF, 2024, L2_01). These results empirically validate the role of DERs as dispatchable, reliable capacity resources under a coordinated DSO–IESO framework.

Complementing this pilot-scale evidence, the GTA North IRRP embeds DERs and eDSM directly into regional planning assumptions, identifying 35–40 MW of combined DG and eDSM peak reduction, a screened-in DER/NWA opportunity of 60–80 MW across key sub-regions, and a long-term residual need of ~600 MW (summer) and ~1,200 MW (winter) (IESO & Technical Working Group, 2025, L2_03). This explicit integration demonstrates that DERs are formally incorporated into Ontario's load-modifying and capacity-alleviating strategies.

Level 1 sources provide global and national context for PV adoption, whereas Level 2 sources provide Ontario-specific evidence of DER performance and planning integration. When analyzed together, they show substantial alignment: economic viability (Level 1) pairs with operational feasibility and system readiness (Level 2).

7.3 Regulatory Alignment and Market Design Convergence

The OEB's Framework for Energy Innovation introduces Ontario's first formal valuation and cost-recovery architecture for DER-based NWAs (OEB, 2023, L2_02). The BCA Framework establishes the "costs should follow benefits" principle, uses multi-layer tests, standardizes DER valuation across distributors, clarifies cost recovery through deferral accounts and prudence review, and enables pilot incentive mechanisms (OEB, 2023, L2_02).

This regulatory framework aligns with the York Region Demonstration, which provided the empirical basis for understanding DER pricing, availability, and performance, and with the GTA North IRRP, which operationalizes this policy direction by formally screening for NWAs (IESO & ICF, 2024, L2_01; IESO & Technical Working Group, 2025, L2_03). The resulting pipeline—pilot → regulatory framework → regional planning integration—indicates that PV and DER adoption is increasingly governed by standardized, evidence-based economic and planning mechanisms.

7.4 System Needs, Resource Potential, and Locational Value

Level 1 sources highlight strong national and provincial growth pathways for solar, declining costs, and mid-term investment requirements (NREL, 2024, L1_01; CanREA & Dunsky Energy + Climate, 2025, L1_02; Smart Prosperity Institute, 2021, L1_03). Level 2 sources refine this broad potential into location-specific insights, detailing GTA North's rising summer and winter peaks, declining but still relevant DG contributions, 60–80 MW of near-term DER/eDSM deferral potential, and long-term residual needs (IESO & Technical Working Group, 2025, L2_03).

Local energy prices from the York Region NWA Demonstration—$1.81/kWh and $1.19/kWh average accepted offers, with a $2.00/kWh ceiling—illustrate the high locational value of DERs in constraint zones (IESO & ICF, 2024, L2_01). While not intended as permanent price signals, they underscore the economic significance of DER deployment in targeted areas. In this context, PV's approximately 28%

summer peak coincidence remains a key driver of its locational capacity value despite limited winter output.

7.5 Overall Synthesis

When the Level 1 and Level 2 sources are combined, they present a coherent adoption narrative: Ontario's evolving system needs, rapid electrification, and rising locational constraints are occurring in parallel with falling PV costs, proven DER operational readiness, maturing planning tools, and expanding regulatory pathways for NWAs. The intersection of these trends suggests a high-potential environment for expanding solar-driven DER portfolios, particularly in rapidly growing regions such as York Region.

The evidence across all sources therefore supports the conclusion that Ontario is structurally prepared, and increasingly economically incentivized to integrate solar PV and DERs as core components of its long-term capacity strategy, not merely as supplementary alternatives. Across these sources, key findings converge on PV's quantified contribution to peak capacity: a summer-peak coincidence of approximately 28% and a dependable-capacity contribution in the 15–30% range, depending on siting and configuration (Dunsky Energy Consulting, 2022, L1_04; National Renewable Energy Laboratory, 2024, L1_06). Together with empirically demonstrated DER portfolio performance in York Region, these characteristics make PV a credible and quantifiable component of distribution-level deferral strategies when combined with storage and demand response (IESO & ICF, 2024, L2_01).

---

8. Policy Implications and Conclusion

Ontario's electricity sector is entering a decade defined by simultaneous electrification-driven demand growth, aging infrastructure, and increasing policy commitment to renewable energy and local flexibility resources. The evidence across the Level 1 (PV cost and market outlook) and Level 2 (DER/NWA, regulatory, and regional planning) sources demonstrates that solar PV and distributed energy resources are positioned to play a structurally important role in meeting provincial reliability and capacity needs. The convergence of cost declines, operational feasibility, and regulatory modernization produces a clear set of policy implications.

8.1 Policy Implications for Ontario's Electricity System

8.1.1 Declining PV Costs and Strengthened Investment Signals

Across NREL ATB, CanREA 2025, and Smart Prosperity, solar PV costs decline steadily through 2035 and 2050, with utility-scale LCOE falling as low as $20–26/MWh by mid-century (NREL, 2024, L1_01; CanREA & Dunsky Energy + Climate, 2025, L1_02; Smart Prosperity Institute, 2021, L1_03). These cost trajectories place solar in direct competition with conventional generation and make it increasingly attractive as a component of provincial supply portfolios and local capacity planning. Nationally, this aligns with Canada Energy Regulator projections showing PV capacity expanding to roughly 20 GW by 2050 under the Evolving Scenario.

8.1.2 DER Performance Validates Non-Wires Opportunities

The York Region NWA Demonstration provides empirical support for the reliable operation of DERs, including high availability and performance accuracy, multiple activation events, 366 MWh of delivered

energy, and near-full subscription in consecutive capacity auctions (IESO & ICF, 2024, L2_01). These results confirm that DERs can produce measurable peak relief under Ontario-specific conditions and under a coordinated IESO–DSO market structure.

8.1.3 Regulatory Framework Now Enables DER Deployment at Scale

The OEB FEI BCA Framework establishes the first standardized provincial methodology for evaluating DER solutions, built around "costs should follow benefits," multi-test analysis, standardized valuation methods, clarified cost-recovery pathways, and pilot incentive mechanisms (OEB, 2023, L2_02). This regulatory clarity enables distributors to consider DERs as legitimate, rate-regulated solutions for deferring or avoiding traditional capital projects and resolves historical disincentives by clarifying recovery of DER-related OM&A and enabling experimental remuneration models (OEB, 2023, L2_02).

8.1.4 Regional Planning Integrates DERs as Standard Planning Resources

The GTA North IRRP institutionalizes DERs and eDSM within regional planning, identifying 35–40 MW of peak reduction by early 2030s, a screened-in NWA opportunity of 60–80 MW, and long-term unserved capacity needs of ~600 MW (summer) and ~1,200 MW (winter) despite committed wires reinforcements (IESO & Technical Working Group, 2025, L2_03). These values demonstrate that DERs are formally recognized components of Ontario's IRRP process.

8.1.5 Locational Value Reinforces the Case for DER Integration

Locational price signals from the York Region Demonstration—average accepted offers of $1.81/kWh and $1.19/kWh with a $2.00/kWh ceiling—far exceed Ontario's wholesale LMPs, demonstrating that DERs provide disproportionate value in constrained feeder and transformer zones (IESO & ICF, 2024, L2_01).

8.2 Strategic Opportunities for Ontario

Integrating Level 1 and Level 2 evidence points to several policy and planning opportunities:

- **Strengthen local capacity markets.** York Region demonstrates that competitive DER procurement can achieve price convergence with wholesale capacity markets while providing additional locational value (IESO & ICF, 2024, L2_01).
- **Implement multi-year DER commitments.** Participant feedback identifies multi-year contracts as a key enabler of investment certainty (IESO & ICF, 2024, L2_01).
- **Expand standardized DER baseline and M&V protocols.** Both FEI and NWA evaluation findings emphasize the need for consistent telemetry, dispatch verification, and baseline methodologies to scale DER participation (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02).
- **Use regional planning to prioritize high-value nodes.** The IRRP identifies where DER potential is most valuable (e.g., Markham, Vaughan, Richmond Hill), enabling targeted procurement aligned with system needs (IESO & Technical Working Group, 2025, L2_03).
- **Align PV deployment with winter reliability strategy.** Given the IRRP's forecast of 5,700 MW winter peaks by 2043, PV's economic competitiveness should be paired with complementary local solutions to address emerging winter constraints (NREL, 2024, L1_01; IESO & Technical Working Group, 2025, L2_03).

8.3 Conclusion

Taken together, the Level 1 and Level 2 sources present a consistent, evidence-based picture of how Ontario can leverage solar PV and distributed energy resources to address emergent reliability needs, manage infrastructure investment, and accelerate cost-effective electrification.

Level 1 sources demonstrate that PV is becoming increasingly cost-competitive, with strong national and continental market momentum and stable long-term cost declines (NREL, 2024, L1_01; CanREA & Dunsky Energy + Climate, 2025, L1_02; Smart Prosperity Institute, 2021, L1_03).

Level 2 sources show that Ontario has already proven the operational viability of DERs, established regulatory mechanisms to value and recover DER-related costs, and integrated DERs into formal regional planning processes (IESO & ICF, 2024, L2_01; OEB, 2023, L2_02; IESO & Technical Working Group, 2025, L2_03).

This convergence signals a decisive transition: Ontario is no longer in a pilot or exploratory phase. It possesses the economic conditions, institutional mechanisms, and empirical evidence necessary to systematically deploy solar PV and DER-based non-wires alternatives as core elements of its future grid strategy. Solar PV's declining costs, DERs' demonstrated performance, and Ontario's maturing regulatory frameworks collectively position distributed and renewable resources not as optional supplements, but as primary tools for managing rising peak demand and optimizing long-term grid investments. At the same time, long-term system reliability planning must continue to account for winter peak shortfalls, where PV output is comparatively lower and complementary resources will remain essential.

For Deloitte and other institutional stakeholders, the synthesis across Level 1 and Level 2 sources indicates that PV now represents a scalable and quantifiable NWA technology for near-term deployment in Ontario. It provides a defensible framework for integrating PV into cost-deferral, emissions-reduction, and grid-modernization strategies that align with regulatory neutrality, economic efficiency, and customer value (OEB, 2023, L2_02; IESO & ICF, 2024, L2_01; IESO & Technical Working Group, 2025, L2_03).

Glossary of Key Terms Used in the Report

Avoided Costs

Costs that the electricity system does not have to incur because a non-wires alternative (e.g., PV, storage, demand response) delays or replaces traditional infrastructure like transformers, stations, or transmission lines.

Availability (DER Availability)

A measure of how often DERs were ready and capable of responding during called events. In the York Region project, availability averaged **83%**, meaning DERs were reliably "on-call."

Benefit-Cost Analysis (BCA)

A standardized regulatory method used by the Ontario Energy Board (OEB) to evaluate whether DER or PV-based solutions deliver net benefits compared with traditional utility infrastructure.

Capacity Factor

The percentage of a power plant's theoretical maximum output that it actually produces over time. PV's capacity factor usually ranges from **26–30%** depending on technology and location.

Capital Expenditures (CapEx)

Upfront investment cost of building a PV system or grid asset, measured in $/kW. A key driver in Levelized Cost of Energy (LCOE).

Coincidence Factor / Peak-Coincidence Factor

The share of a resource's output that aligns with the system's highest-demand hours (peak). Rooftop PV's summer coincidence factor is **~28%**.

DER (Distributed Energy Resource)

Smaller, decentralized electricity resources located close to customers—examples: rooftop solar, energy storage, EV chargers, CHP, or industrial demand response—capable of supporting grid needs.

DER Host Impacts

The local system effects that arise where a DER is physically connected, such as voltage impacts, protection settings, or backfeed conditions.

Distributed Generation (DG)

Electricity generation located on the distribution network or behind customer meters. In Ontario this primarily refers to legacy **FIT/microFIT PV** installations.

Distribution System Operator (DSO)

A future-looking role for local distribution companies where they actively coordinate DERs, run local markets, and manage local grid flexibility (tested in the York Region pilot).

Electrification

Growth in electricity demand driven by the transition from fossil fuels to electric solutions—particularly EVs and electric heating—creating new system peaks in Ontario.

Extreme-Weather Coincident Peak

Ontario's formal planning metric representing peak load under extreme heat or cold conditions. PV contributes mainly to summer peaks but little to emerging winter peaks.

Feed-In Tariff (FIT) / microFIT

Ontario's legacy program that guaranteed fixed prices for solar generation, creating the province's first major wave of distributed PV. These contracts are now expiring.

Framework for Energy Innovation (FEI)

The OEB's regulatory framework that provides valuation rules, incentives, and BCA methods for utilities when evaluating DERs.

Integrated Regional Resource Plan (IRRP)

The IESO's long-term planning process for specific geographic areas. The **GTA North IRRP (2025)** is central to this report.

Intermittency

The variable nature of renewable resources like PV that depends on sunlight availability and therefore cannot produce electricity continuously.

Levelized Cost of Energy (LCOE)

A metric that captures the lifetime cost of generating electricity from a resource, expressed as $/MWh. Utility-scale solar is projected to reach **$20–26/MWh** by mid-century.

Local Capacity Auction

A market mechanism piloted in the York Region NWA Demonstration where DERs bid to provide local peak-capacity services, priced in **$/kW-day**.

Local Energy Auction

A companion market in the York Region pilot where DERs are paid for actual dispatch energy delivered during constrained periods (up to $2.00/kWh in the pilot).

Load-Modifying Resource

Any resource—PV, DSM, DG—that reduces net system demand seen by the grid operator.

MW-Year (Per-MW-Year Value)

A standardized way to compare the annual economic benefit or cost of 1 MW of capacity from any technology (DER or wires).

Net Extreme-Weather Peak

The IESO's calculation of peak demand after accounting for load-modifying resources such as PV and DSM.

Non-Wires Alternative (NWA)

Any non-traditional solution (PV, storage, demand response, eDSM) that can defer or avoid investments in conventional "wires" infrastructure.

NWA Screening Tool

The IESO's formal methodology to evaluate if a capacity problem can be cost-effectively solved with DERs instead of transformers, feeders, or transmission expansions.

Ontario Energy Board (OEB)

The regulator responsible for distribution utility oversight, cost-recovery rules, and DER valuation frameworks in Ontario.

Peak Demand / Peak Load

The maximum electricity demand in a region during a specific period. Ontario is transitioning from a summer-peaking to a winter-peaking system due to electrification.

Performance Accuracy

How closely a DER delivers the capacity it commits relative to what was expected. The York Region pilot achieved **85%** accuracy.

Photovoltaics (PV)

Solar technology that converts sunlight into electricity. The report focuses on its role as a DER and potential NWA resource.

Resource Adequacy

System operator assessment of whether sufficient capacity exists to meet future electricity demand reliably.

Station Upgrades (e.g., Kirchberg TS, Kleinburg TS)

Large capital projects that expand transmission or transformer station capacity. NWAs aim to defer or downsize these.

Summer vs. Winter Peaks

Ontario is shifting from summer-dominant peaks (driven by AC) to winter-dominant peaks (driven by heat pumps and building electrification). PV aligns well with summer peaks but not winter peaks.

Total-DSO Model

An operational model tested in the York Region demonstration where the local utility coordinates DER operations, markets, and dispatch under a distribution-level market framework.

Transmission Deferral

The avoided cost of delaying transmission line or terminal station expansion thanks to DERs providing local capacity.

Utility Remuneration

Mechanisms by which utilities recover costs and earn returns. The FEI introduces new incentive structures to encourage DER adoption (e.g., shared savings, scorecards, margin-on-payments).

References (by Source ID)

L1_01 National Renewable Energy Laboratory (NREL). (2024). *Annual Technology Baseline (ATB) 2024 – Photovoltaic cost and performance data*. U.S. Department of Energy. https://atb.nrel.gov/

L1_02 Canadian Renewable Energy Association (CanREA), & Dunsky Energy + Climate. (2025). *Canadian Renewable Energy Association Market Outlook 2025 – Executive summary*. https://renewablesassociation.ca/

L1_03 Smart Prosperity Institute, University of Ottawa. (2021). *Material needs and end-of-life resource flow implications under Canada's climate change objectives and data gaps (Part 1: Solar and wind)*. https://institute.smartprosperity.ca/sites/default/files/Working_Paper_Solar_Energy_01.pdf

L1_04 Dunsky Energy Consulting. (2022). *Distributed energy resources potential study – Volume I*. Report prepared for the Independent Electricity System Operator (IESO). https://dunsky.com/wp-content/uploads/DER-potential-study-IESO-Dunsky-Vol1.pdf

L1_05 Gaucher-Loksts, E., & Pelland, S. (2024). *Assessing the photovoltaic potential of the Canadian building stock*. Natural Resources Canada (CanmetENERGY). https://natural-resources.canada.ca/sites/nrcan/files/canmetenergy/files/pdf/PV_potential_summary_report_appendix-EN.pdf

L1_06 National Renewable Energy Laboratory (NREL). (2024). *A 2023 perspective: The value of hybridization*. https://docs.nrel.gov/docs/fy24osti/87824.pdf

L1_07 Independent Electricity System Operator (IESO). (2023). *Integrated regional resource plans: Guide to assessing non-wires alternatives*. https://www.ieso.ca/-/media/Files/IESO/Document-Library/regional-planning/IRRP-NWA-Process-Guidelines.pdf

L1_08 Ontario Energy Board (OEB). (2025). *Distributed energy resources connection procedures (Version 2)*. https://www.oeb.ca/sites/default/files/DER_Connection_Procedures_V2.pdf

L2_01 Independent Electricity System Operator (IESO), & ICF. (2024). *IESO York Region non-wires alternatives demonstration – Final evaluation report*. https://www.ieso.ca/en/Sector-Participants/Engagement-Initiatives/Engagements/IESO-York-Region-Non-Wires-Alternatives-Demonstration-Project

L2_02 Ontario Energy Board (OEB). (2023). *Framework for energy innovation: Setting a path forward for DER integration*. https://www.oeb.ca/sites/default/files/FEI-Report-20230130.pdf

L2_03 Independent Electricity System Operator (IESO), & Technical Working Group. (2025). *GTA North (York Region) integrated regional resource plan*. https://www.ieso.ca/-/media/Files/IESO/Document-Library/regional-planning/York/gta-north-York-IRRP-Report.pdf